Compiled Problems – Intermediate Division

1. **01-02 C1 Computer Number Systems**

   Convert BED from hexadecimal to octal.

2. **01-02 C1 Computer Number Systems**

   Solve for X_{16}: $X_{16} = 24415_8 + 56712_8$

3. **01-02 C2 Computer Number Systems**

   Solve for X_2: $X_2 = A12_{16} - 567_8$

4. **02-03 C1 Computer Number Systems**

   How many 1's are in the binary representation of the sum of $24_8 + 372_{16}$?

5. **02-03 C1 Computer Number Systems**

   Solve for X_8: $2_8 X_8 = 4_{16} * AB_{16}$

6. **02-03 C2 Computer Number Systems**

   What is the octal value of $(764_8)^2$?

7. **03-04 C1 Computer Number Systems**

   Evaluate each expression and convert the answer to binary.
   Ignoring leading zeros, which binary answers contain more 1's than 0's?
   List all.

   A. $42_8 - 12_8$ B. $3A_{16} + 10B_{16}$ C. $10001_2 + 11101_2$ D. $A_{16} + 10_8 + 10_2$

8. **03-04 C1 Computer Number Systems**

   Solve for X_{10}: $1000_2 * X_{10} + A_{16} = 33_8 + 17_{16}$

9. **03-04 C2 Computer Number Systems**

   Convert the following hex value to a base 10 value.

   $0.3A_{16} = X_{10}$

10. 04-05 C1 Computer Number Systems

Solve for X_{16}: $327_8 + X_{16} = 1101110110_2$

11. 04-05 C1 Computer Number Systems

Evaluate the following and express the answer in hexadecimal.

$$213_8 + 10110_2 - 2A_{16}$$

12. 04-05 C2 Computer Number Systems

Multiply and express the answer in octal: $FF_{16} * 10001_2$

13. 05-06 C1 Computer Number Systems

Which of the following is the largest?

A. 53_{16} B. 1010001_2 C. 124_8 D. 82_{10}

14. 05-06 C1 Computer Number Systems

Evaluate the following and express the answer in hexadecimal.

$$213_8 + 10110_2 - 2A_{16}$$

15. 05-06 C2 Computer Number Systems

Place the following numbers in descending order with respect to the number of 1's in their binary representation. Using the multiple-choice letters, place the letters in the provided boxes with the letter representing the number with the largest number of 1's in the top box.

A. 231_{16} B. 73_8 C. 10111011_2 D. 84_{10}

16. 06-07 C1 Computer Number Systems

Convert from hexadecimal to octal: $A2B8C4D6_{16}$

17. 06-07 C1 Computer Number Systems

A palindrome is a number like 1221 that reads the same from left to right as it does from right to left. Convert the numbers from 10_{10} to 100_{10} to octal. How many octal palindromes exist in that interval?

18. 06-07 C1 Computer Number Systems

What is the positive difference between the number of 1's in the binary representation of the following two values:

$A37E_{16}$ and 67541_8

19. 07-08 C1 Computer Number Systems

Which of the following octal numbers has the most 1's in its binary representation?

A) 155555_8 B) 133337_8 C) 145547_8 D) 156464_8 E) 171442_8

20. 07-08 C1 Computer Number Systems

Evaluate and express the answer in binary: $243_8 + 91_{16} - 82_{16} / 1101_2$

21. 07-08 C2 Computer Number Systems

Solve for X_{16}: $1F_{16} * X_{16} + 476_8 = 1010110010_2$

22. 08-09 C1 Computer Number Systems

How many more 1's are there in the binary representation of $FE94A_{16}$ than in the binary representation of 7534201_8?

23. 08-09 C1 Computer Number Systems

Evaluate and express the answer in hexadecimal:

$13 + F4_{16} - 23_8 + 10101_2$

24. 08-09 C2 Computer Number Systems

Solve for X and express the answer in hexadecimal digits: $4_{16} * X - E8_{16} = C8_{16}$

25. 09-10 C1 Computer Number Systems

Evaluate the following: $X_{16} = 23_8 * 1101_2 + A1_{16} - 111110_2 / 1F_{16}$

26. 09-10 C1 Computer Number Systems

Solve for X_8 in the following: $2A_{16} + X_8 = 101110_2 / 27_8 + 250_{10}$

27. 09-10 C2 Computer Number Systems

Given a binary number, add it to its digits reversed. If the resulting sum is a not a palindrome (same read forwards and backwards), then repeat the process until a palindrome is achieved. What is the palindrome? The starting binary number is 10011.

28. 10-11 C1 Computer Number Systems

Evaluate the following and express the answer as a hexadecimal:

$$176_8 + 10110_2 / B_{16} + 54_{16} * 1011_2$$

29. 10-11 C1 Computer Number Systems

How many 1's are in the binary representation of the positive difference between following the largest and smallest numbers?

$$111_8 \quad 3B_{16} \quad 111110_2 \quad 112_8 \quad 58_{16}$$

30. 11-12 C1 Computer Number Systems

Which hexadecimal number has the most 1's when written in binary notation?

$$576_{16} \qquad B2C_{16} \qquad 28F_{16} \qquad CCF_{16}$$

31. 11-12 C1 Computer Number Systems

Evaluate the following expression and express the answer in octal:

$$13_8 + A3_{16} - 1101_2$$

32. 12-13 C1 Computer Number Systems

Which number is the largest?

$$2A3_{16} \qquad 1243_8 \qquad 678_{10} \qquad 1010100101_2$$

33. 12-13 C1 Computer Number Systems

Evaluate the following expression and express the answer in octal: $101101_2 * B7_{16}$

34. 13-14 C1 Computer Number Systems

Find the average of the following. Express the answer in octal.

110111_2 63_{10} $3A_{16}$ 124_8

35. 13-14 C1 Computer Number Systems

Which of the following has the most 1's in its binary representation?

178_{16} 567_8 101110110_2 565_8 377_{10}

36. 14-15 C1 Computer Number Systems

Which expression evaluates to the larger number?
Express answer as a hexadecimal number.

$15_8 + B1C_{16} - 100110_2$ or $AB8_{16} + 127_8 + 1000_2$

37. 14-15 C1 Computer Number Systems

What is the next hexadecimal number after $A47_{16}$ to have the substring 10101 appear in its binary representation?

38. 15-16 C1 Computer Number Systems

Which expression evaluates with the most 1's when the answer is expressed in binary?

A) $25_8 + 1C_{16}$ B) $A0_{16} - 100110_2$ C) $B8_{16} + 127_{10}$ D) $111_8 + 1000_2$

39. 15-16 C1 Computer Number Systems

What is the mean (average) of the following numbers expressed in octal?

101010_2 132_8 33_{16} 1010100_2 44_{16}